

ED 111 – Engineering Mathematics I

Academic year: 2025-26

Semester: Autumn 25

Instructor Nabin

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Room 4212 Faculty Block 4

Credit Structure : (L-T-P-Cr): (3-1-0-4)

Course Code: ED 111

Program/ Semester: B.Tech, Semester I

Category: Core

Prerequisite: None

Tentative Course Content

Calculus: Limit, continuity, differentiation, application of derivatives, integration, definite integrals, method of substitution, integration by parts, application to finding area and volume, double and triple integrals, improper integrals.

Matrix:

Matrix addition, subtraction, multiplication, determinants, rank of a matrix, eigenvalues, eigenvectors and applications.

Vectors: Vector algebra, products of vectors, vector triple product, vector equations,

Differentiation and integration of vectors, vector calculus.

Differential Equations:

Formation and solution of ODE, solution of first order ODEs, Linear inhomogeneous second order ODEs, free and forced oscillators and applications.

Formation and solution of PDEs, Second order linear PDE, Diffusion equation and applications.

Complex Numbers:

Complex numbers and its properties, polar representation, Euler's formula, exponential form.

Numerical methods:

Newton's method for finding roots, numerical integration methods, finite element method.

Expected Outcome

The students after completing the course will get a basic overview of calculus, linear algebra, differential equations, complex analysis and numerical analysis. Students will get an insight of fundamentals of these topics and their vast applications. In some of the tutorials we will use some computer programming to visualize and understand the solutions. All these outcomes will be assessed with homework, quiz and exams.

At the end of the course the students will be able to
- Understand and define limit, continuity, differentiability of a function at a point. - evaluate limits using the definition and using algebraic properties of limits . - compute indefinite and definite integration and its application. - interpret the area bounded by curves, volume of a solid of revolution as a definite integral etc.
- represent the system of linear equation in matrix and vector form. - solve system of linear equations using matrices. - solving real life problems using matrices - find and understand determinant and linear transformation. - find eigen value and eigen vector of a square matrix and its applications. - understand vector algebra and vector calculus.

- form ODE of first and second order and will learn about the solutions. - understand the concept of a general solution, a particular solution and initial conditions. - draw slope fields by hand and also by using computer. - solving first order ode (linear and nonlinear) using different methods. - solution of second order linear ODE with constant and variable coefficient. - form PDE and learn about the solutions. - learn about the applications of PDEs
- understand the idea of complex numbers - represent complex numbers in its polar form - find complex differentiation and integrations
- find the roots using different methods - numerically integrate a function

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X				X			X

Textbook

The following textbooks will be helpful. We will provide many additional materials such as lecture notes etc. during the course. There is no need to purchase any book.

1. Thomas' Calculus
2. Calculus by Gilbert Strang
3. Advanced Engineering Mathematics, by Erwin Kreyszc
4. Advance Engineering Mathematics by Jain and Iyenger
5. Engineering Mathematics by Anthony Croft, Robert Davison and Martin Hargreaves
6. Calculus by James Stewart

Mark Distribution (Tentative) / Grading Policy

First in sem-25%

Second in sem- 25%

End sem -40 %

Attendance &

Quizes-10%

Tutorials

This is a class with your Instructor/TA during our regularly scheduled tutorial time. These will be used as a guided homework session where you work together with your instructor/TA on the homework sets, and to answer any questions you may have related to the course.

Course Resources

Lecture notes.

Lecture notes/ class notes will be posted and made available for students after the lecture in the google class room.

Tentative Lecture Schedule

Sl. No.	Description	No. of Lectures
1	Calculus Limit, continuity, differentiation, application of derivatives, integration, definite integrals, method of substitution, integration by parts, application to finding area and volume, double and triple integrals, improper integrals.	10
2	Linear Algebra System of linear equations, matrix, matrix addition, subtraction, multiplication, determinants, rank of a matrix, eigenvalues, eigenvectors and applications.	6
3	Vectors: Vector algebra, products of vectors, vector triple product, vector equations, differentiation and integration of vectors, vector calculus.	7
4	Differential Equations: Formation and solution of ODE, solution of first order ODEs, Linear inhomogeneous second order ODEs, free and forced oscillators and applications. Formation and solution of PDEs, Second order linear PDE, Diffusion equation and applications.	8
5	Complex Numbers: Complex numbers and its properties. polar representation, Euler's formula, exponential form.	6
6	Numerical methods: Newton's method for finding roots, numerical integration methods, finite element method.	4